



Uliana Hrab

DATA · MACHINE LEARNING · DEVELOPMENT

I am in my fourth year of a BSc in Applied Mathematics, based in Norway. The degree covered a wide range of subjects: machine learning and deep learning, computer vision, blockchain, computational geometry, C++ and some economics. Data work is what I like most, so I am looking mainly for analyst and ML/DL roles, though I am open to other positions. I am ready to learn on the job and to take additional courses for a qualification, with support from NAV.

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LOCATION

Norway

GITHUB

github.com/draurug

SELECTED WORK

| Longer write-ups of each project are on the work page.

Ethereum Fraud Detection

2026

Classified 9,841 Ethereum addresses as fraudulent or legitimate from 50 transaction-derived features. LightGBM reached F1 0.987 on held-out data. Before that, I found and fixed a leak: a naive random split put 5.7% of test rows into the training set through duplicate feature profiles. I made the split group-aware and selected models by cross-validated PR-AUC instead of test score.

Python · scikit-learn · LightGBM · XGBoost · pytest

CrossW

2026

Designed and shipped a themed crossword site, from the product idea to a live deployment. Four content packs and 32 puzzles live, 75 unit tests, roughly 5,400 lines of TypeScript. Answers are compiled to opaque indices at build time, so they never reach the browser in readable form.

Next.js 15 · React 19 · TypeScript · Vitest · Cloudflare Workers

Voronoi Path Planner

2026

Interactive path planner that optimises for clearance instead of distance. It builds a clearance field against the obstacles and binary-searches the largest threshold that keeps start and goal connected. Dijkstra then runs inside that corridor. Comes with a generated figure set and a written report.

C++ · OpenGL · FreeGLUT · NumPy · SciPy

Networked Tic-Tac-Toe and Battleship

2026

Two Qt desktop games in C++. The first is a client for a multiplayer game server: invitation handshake, turn synchronisation and game-over detection, on top of a third-party Boost.Asio transport from the course. In the second, the game rules are entirely free of Qt and the widget only paints.

C++17/20 · Qt · CMake

TECHNICAL SKILLS

LANGUAGES

Python, C++, TypeScript, SQL, Octave/MATLAB

MACHINE LEARNING AND DATA

scikit-learn, LightGBM, XGBoost, pandas, NumPy, SciPy, matplotlib, seaborn, Jupyter, Power BI

TOOLS

PostgreSQL, MySQL, Git, CMake, pytest, Qt, OpenGL, Next.js and React, LaTeX

EDUCATION

BSc Applied Mathematics — Odesa I. I. Mechnikov National University

2023 — present

Currently in the fourth year, at the Institute for Optimal Control and Economic Cybernetics. Course paper on fraud detection.

Introduction programme — Læringssenteret for voksne, Ringerike kommune

May — Dec 2025

Norwegian from A1 to B2, civics, and a work placement.

IT Step Academy — IT-focused secondary programme

2019 — 2022

Programming fundamentals, game development in Unity and C#, 3D modelling in Blender, Arduino.

Odesa Lyceum no. 1 (A. P. Bystrina) — secondary education

2016 — 2023

English specialisation.

EXPERIENCE

Medarbeider — Kirkens Bymisjon Hønefoss

Aug — Nov 2025

Language-practice placement (språkpraksis) through the introduction programme. Worked shifts as part of the team, in Norwegian.

Logistics manager — SmartAuto LLC

May — Aug 2024

Coordinated vehicle logistics. Maintained the operational records the scheduling ran on.

Mentor — IT2School

Dec 2023 — Jan 2025

Taught programming to school students, helped them through their assignments and ran masterclasses.

Deputy chair — faculty student union committee, Odesa I. I. Mechnikov National University

2023 — 2025

Organised events for students at the faculty.

Project lead — Mechnikov student card

2023 — 2025

Led the student card project at the university. Member of AtomSpace.

LANGUAGES

- **Russian** and **Ukrainian** — native
- **English** — C1, certified by the London School of English in Ukraine (2025)
- **Norwegian** — B2, confirmed by Norskprøven; course completed at Læringssenteret for voksne i Ringerike kommune, May 2026

CERTIFICATES

- English C1 — London School of English in Ukraine, 2025
- Working with Data in Python — DataCamp, 2024
- Data Visualisation in Power BI — DataCamp, 2024
- Data Analysis in Google Sheets — DataCamp, 2024
- C++ Intermediate — Sololearn, 2023
- C++ Basics — Coursera, 2023

REFERENCES

Available on request.